

MOHAMMED AQEEL ISMAIL

🏠 Pietermaritzburg, South Africa | 📞 Cell no.: 081 711 5093 | ✉️ aqeelismail06@gmail.com | 🌐

www.linkedin.com/in/aqeel06 | 🐙 GitHub: <https://github.com/tomRiddle-the1st> | 📁 Portfolio: tomriddle-the1st.github.io

Professional Summary:

Computer Science graduate with a BSc in Computer Science and Information Technology and a BSc Honours in Computer Science. Possesses strong foundations in Python, Java, C++, SQL, data structures, algorithms, artificial intelligence, machine learning, and software development. Experienced as a Programming Demonstrator, supporting undergraduate students in programming and problem-solving while developing strong communication and mentoring skills. Passionate about teaching, academic development, and helping students build confidence in computing concepts and practical programming.

Education:

BSc Honours (Computer Science), University of KwaZulu-Natal, Pietermaritzburg, South Africa

Mar 2025 – Graduation Date: 14th May 2026

BSc Computer Science and Information Technology, University of KwaZulu-Natal, Pietermaritzburg, South Africa

Mar 2022 – Graduation date: 15th May 2025

National Senior Certificate, Raisethorpe Secondary School, Pietermaritzburg, South Africa

2016 – 2020

Achieved 4 distinctions with an A Aggregate pass

Relevant Coursework:

- | | | | | | | |
|------------------------|-----------------------------------|------------------------------|-------------------------------|--|--|---------------------------------|
| • Computer Programming | • Foundations of Computer Science | • System Analysis and Design | • Object-Oriented Programming | • Database and Programming | • Computer Organisation and Architecture | • Data Structures |
| • Computer Systems | • Network and Database Management | • Theory of Computation | • Artificial Intelligence | • Image Processing and Computer Vision | • Machine Learning | • Applied System Implementation |

Technical Skills:

Programming Languages: HTML, CSS, Python, Java, C++, SQL, C#

Frameworks and Tools: Wing, Notepad++, JGrasp, Eclipse, Clion, SQLSMS19, Visual Studio, Google Colab, IntelliJ IDEA, ASP.NET, PowerBI, Scikit-Learn, Pandas, NumPy

Core Competencies: OOP, Web Development, Data Structures and Algorithms, Database Management, Problem Solving, Team Collaboration, Debugging, AI, ML Basics

Experience:

Programming Demonstrator (Part-time), University of KwaZulu-Natal, Pietermaritzburg, South Africa

Feb 2024- Oct 2024 | Aug 2025- Oct 2025

Tutored first-year and second year students in mastering Python, Java and Data Structures during practical lab sessions. Guided students through debugging, code design and algorithm implementation. Delivered structured lab sessions improving programming comprehension and performance. Provided individual academic support to strengthen foundational coding skills.

Key Projects:

Point of Sales system for Townbush Pharmacy (Third year group project- 2024)

Tech: C#, ASP.NET, SQL Server

- Developed a secure full-stack POS system with authentication, cart functionality, and transaction processing.
- Designed and implemented a normalized SQL database for inventory and sales management.
- Optimised database queries to improve system performance and efficiency.

Simple Biology Quiz and Word Search Game (Third year group project- 2024)

Tech: C++

- Led development of an interactive educational game with difficulty levels and timers.

- Implemented modular and scalable logic for quiz generation and gameplay mechanics.

Network Traffic Classification of 5G Networks for QoS Improvement (Honours Research- 2025)

- Developed a supervised machine learning model to classify network traffic using flow-based features.
- Improved Quality of Service (QoS) decision-making through accurate traffic classification.
- Conducted data preprocessing, feature extraction, and model evaluation.
- Delivered academic presentations within the School.

Achievements:

- Research paper accepted for publication in the Indonesian Journal of Computer Science. Available online on: [Preprints.org](#).
- Successfully mentored and supported over 100+ students in programming modules.
- Developed multiple real-world academic software systems and ML models.

Strengths:

Problem-solving ability
Basic programming fundamentals
Willingness to learn & adapt
Communication and teamwork

References:

Mr P. T. Gounden – Principal, Raisethorpe Secondary School
Contact: 071 345 1632

Mrs Rosanne Els – Lecturer, University of KwaZulu-Natal
Contact: 065 335 5686

Mr Jean Vincent Fonou Dombu – Lecturer, University of KwaZulu-Natal
Contact: 033 260 5644

Mr Okuthe Paul Kogeda – Lecturer & Honours Supervisor, University of KwaZulu-Natal
Contact: 082 471 2799